RAJALAKSHMI ENGINEERING COLLEGE

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T2

Started on	Friday, 14 August 2026, 2:29 PM
State	Finished
Completed on	Friday, 11 September 2026, 1:47 PM
Time taken	27 days 23 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than `⌊n / 2⌋` times. You may assume that the majority element always exists in the array.

Example 1:

```
Input: nums = [3,2,3]
Output: 3
```

Example 2:

```
Input: nums = [2,2,1,1,1,2,2]
Output: 2
```

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 10^4`
- `-2^31 <= nums[i] <= 2^31 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int cnt(int a[],int l,int r,int x){
    int c=0;
    for (int i=l;i<=r;i++){
        if (a[i]==x)
            c++;
    }
    return c;
}
int maj(int a[],int l,int r){
    if (l==r){
        return a[l];
    }
    int m=(l+r)/2;
    int x=maj(a,l,m);
    int y=maj(a,m+1,r);
    if (x==y){
        return x;
    }
    int c1=cnt(a,l,r,x);
    int c2=cnt(a,l,r,x);
    return (c1>c2)?x:y;
}
int main(){
    int n;
    scanf("%d",&n);
```

```
27      int a[n];
28      for (int i=0;i<n;i++){
29          scanf("%d",&a[i]);
30      }
31      printf("%d",maj(a,0,n-1));
32  }
```

	Input	Expected	Got	
✔	3 3 2 3	3	3	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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